

VEDANT KAPIL

AI Engineer | Machine Learning | Generative AI | Python Developer

India | Portfolio | GitHub | LinkedIn | Email

PROFESSIONAL SUMMARY

AI Engineer with hands-on project experience building applications using Large Language Models, Retrieval-Augmented Generation (RAG), Computer Vision, and Machine Learning. Skilled in Python, LangChain, FastAPI, and data analytics, with a track record of shipping RAG systems, AI agents, and client-facing web applications. Continuously expanding expertise in AI engineering, backend development, and MLOps.

TECHNICAL SKILLS

Programming: Python, C, C++, SQL

Generative AI: Large Language Models (LLMs), LangChain, LangGraph, RAG, Prompt Engineering, Embeddings, Sentence Transformers, Ollama, ChromaDB, AI Agents

Machine Learning: Supervised Learning, Regression, Classification, Model Evaluation, Data Preprocessing

Data Analytics: NumPy, Pandas, Matplotlib, Scikit-learn, Data Cleaning, Exploratory Data Analysis, Feature Engineering, Data Visualization

Computer Vision: YOLO, OpenCV, Roboflow

Backend & Web: FastAPI, REST APIs, JSON, HTTP APIs, Next.js, React, TypeScript

Developer Tools: Git, GitHub, VS Code, Linux, Vercel

Currently Learning: Docker, AWS, PyTorch, Deep Learning

PROJECTS

PDF RAG Chatbot | *Python, LangChain, ChromaDB, Ollama, Sentence Transformers, Streamlit*

- Built a Retrieval-Augmented Generation chatbot that ingests user-uploaded PDF documents and answers natural-language questions with context-grounded responses.
- Designed a document ingestion pipeline to extract, clean, and chunk PDF text, then converted chunks into embeddings using Sentence Transformers.
- Indexed embeddings in ChromaDB for fast semantic retrieval and generated responses locally through Ollama via LangChain.
- Built an interactive Streamlit interface supporting PDF upload, real-time query, and streaming chat responses.

Sindhu Construction — Freelance Website | *Next.js, React, TypeScript, Tailwind CSS, Vercel*

- Designed and developed a full marketing website for a Bengaluru-based construction and interior design company as a freelance client project.
- Built a filterable project portfolio/monograph section and an interactive before/after renovation showcase.
- Implemented a commission guide and cost-estimation page along with an FAQ and lead-inquiry contact section.
- Deployed on Vercel with a responsive, animation-driven UI and SEO-optimized metadata for organic discovery.

AI Portfolio Website | *React, TypeScript, GitHub API, GitHub Pages, REST APIs*

- Designed and developed a modern portfolio website using React and TypeScript.
- Integrated the GitHub API to automatically display repositories, keeping project listings up to date without manual edits.
- Implemented an admin dashboard enabling portfolio content updates through GitHub using Personal Access Tokens.
- Configured GitHub Pages deployment to automatically publish updates whenever repository content changes.

Amazon Product RAG | *Python, LangChain, ChromaDB, Ollama, Sentence Transformers, RAG*

- Developed a Retrieval-Augmented Generation (RAG) application capable of answering product-related questions.
- Converted product data into embeddings using Sentence Transformers for semantic search.
- Stored and indexed embeddings in ChromaDB for efficient retrieval.
- Retrieved relevant context using LangChain and generated responses locally using Ollama.

AI Chatbot | *Python, LangChain, Ollama, Streamlit*

- Built a conversational AI chatbot using LangChain and Ollama with local response generation.
- Implemented streaming responses and maintained chat history for continuous conversation context.

- Created an interactive user interface using Streamlit.

Traffic Detection System | *YOLO, OpenCV, Roboflow, Python*

- Developed a real-time object detection system using YOLO for identifying traffic elements.
- Trained and evaluated the detection model using Roboflow datasets.
- Applied OpenCV for image processing and video frame handling.

Data Analytics Projects | *Python, Pandas, NumPy, Scikit-learn, Matplotlib*

- Performed exploratory data analysis (EDA) to uncover trends and data quality issues.
- Built machine learning models using Scikit-learn for regression and classification tasks.
- Applied data cleaning and preprocessing techniques using Pandas and NumPy.
- Created visualizations using Matplotlib to communicate insights clearly.

EDUCATION

Bachelor of Engineering (Computer Engineering)

St. John College of Engineering and Management | 2024 – 2028

Higher Secondary (Class XII) – Science (PCMB)

Motherland Public School (CBSE) | 2022 | 80%

Secondary (Class X)

Motherland Public School (CBSE) | 2020 | 76%

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI & Stanford University (Coursera)
- Introduction to Large Language Models — Google Cloud (Coursera)
- Google AI Essentials — Google
- CS50's Introduction to Programming with Python — Harvard University
- OpenAI Academy: Agents and Workflows
- JPMorgan Chase & Co. Quantitative Research Job Simulation — Forage
- Deloitte Data Analytics Job Simulation — Forage
- Tata GenAI Powered Data Analytics Job Simulation — Forage
- The Ultimate Job Ready Data Science Course
- C Programming for Beginners — Learn C Language from Scratch